Finance 6110: Corporate Finance
Final Exam

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Name: _____

Section (circle one): 8:30 AM 10:15 AM 1:45 PM 3:30 PM

Instructions:

Please write your name and circle your section number before beginning the exam.

Do **not** turn over this exam or begin working until you are told to do so.

Each multiple-choice question has a single correct answer.

Incorrect answers to true/false and multiple-choice questions receive 0 points.

Please write and **circle final answers clearly**. Ambiguous answers will be marked as incorrect.

Unless otherwise specified, round *final* answers to the nearest decimal place, for percentages this means, e.g., 8.3%.

You have 120 minutes to complete the exam. You must hand in the exam on time.

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Part I: True/False (2 points each, 20 points total)

Instructions: Circle true or false to indicate whether each statement is true or false.

1. An appealing feature of a conglomerate merger is that entering a new industry can diversify the firm's cash flows, which is typically more difficult for investors to do on their own.	TRUE or FALSE
2. In a hostile takeover, the firm's managers oppose the deal and the acquirer must instead win over shareholders.	TRUE or FALSE
3. The ability to step up the tax basis of the target's assets is generally a desirable feature of an asset purchase.	TRUE or FALSE
4. A poison pill can give existing shareholders the right to buy additional shares at a discount if a hostile bidder acquires a large stake.	TRUE or FALSE
5. If the acquirer believes there is a risk that the target is overvalued, the acquirer would prefer to pay with stock in order to share this risk.	TRUE or FALSE
6. Ambiguity aversion can increase the cost of capital for firms operating in uncertain or poorly understood environments.	TRUE or FALSE
7. Short-selling constraints help ensure that markets remain efficient by making it easier for arbitrageurs to correct mispricing.	TRUE or FALSE
8. According to Modigliani-Miller Proposition 1, increasing leverage can either increase or decrease total firm value, depending on how the value of equity changes in response to higher leverage.	TRUE or FALSE

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9. Counterparty risk in futures markets is reduced through daily settlement (marking to market).	TRUE or FALSE
10. Whether an option is a call or put option, if it is in-the-money, it is profitable to exercise the option.	TRUE or FALSE

Part II: Multiple Choice (2 points each, 20 points total)

Instructions: Circle the letter of the correct answer for each question.

11. All else equal, which of the following unambiguously **increases** free cash flows to the firm (FCFF)?
    a. Higher capital expenditures
    b. Higher net working capital requirements
    c. Higher depreciation
    d. Higher taxes
    e. None of the above

12. Which of the following should **not** be included in a project's FCFF?
    a. Additional sales cannibalizing an existing product line
    b. Fees paid to consultants to forecast potential costs and revenues that are incremental to the project
    c. Increased maintenance expenses due to new equipment
    d. Additional working capital required by the project

13. All else equal, which of the following leads to a **higher** price-to-earnings (PE) ratio?
    a. Higher risk-free rate
    b. Lower expected earnings growth
    c. Lower payout ratio
    d. Lower equity beta

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14. Which of the following generally applies to an asset purchase but **not** a stock purchase?
    a. The acquirer avoids assuming the target's liabilities
    b. The acquirer inherits the target's net operating losses
    c. Shareholders avoid double taxation
    d. Shareholders decide whether or not to approve the deal, not the target firm's managers
    e. All of the above

15. The surge in the price of the closed-end fund CUBA, which had no investment in the country of Cuba, after US policy announcements about relations with Cuba, is best described as an example of:
    a. Momentum
    b. Overreaction
    c. Underreaction
    d. Loss aversion
    e. Overconfidence

16. Which of the following is **not** implied by the Efficient Markets Hypothesis (EMH)?
    a. Trading on past price trends (technical analysis) will not be profitable on average
    b. Prices respond quickly to new public information
    c. Investors with enough skill and data can consistently outperform the market
    d. Stock prices reflect all available information
    e. None of the above

17. According to Modigliani-Miller Proposition 2, as a firm's leverage increases
    a. Its cost of equity increases
    b. Its WACC remains constant
    c. Its unlevered cost of capital remains constant
    d. All of the above

18. The most common use of forward contracts is to hedge exposure to…
    a. Equity risk
    b. Commodity price risk
    c. Exchange rate risk
    d. Interest rate risk
    e. Real estate risk

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19. Which of the following positions benefits if the spot price **falls**?
    a. Long put
    b. Long call
    c. Long forward
    d. All of the above
    e. None of the above

20. Which of the following is **not** a potential benefit of compensating employees with call options?
    a. Aligns employee incentives to increase firm equity, as call options become more valuable as equity increases in value
    b. Discourages excessive risk-taking, as call options become less valuable as equity volatility increases
    c. Helps conserve cash at the time of issuance
    d. Increases retention through vesting schedules
    e. All of the above

Part III: Multi-Part Problems (77 points total)

Instructions: Enter your answers and show relevant work ***inside the boxes.***

Problem 1: Ditta Project Valuation (20 points)

Ditta, a specialty food company, is considering launching a **miso-based candy bar**. You're tasked with valuing this project.

Cash Flow Forecasts

- The project involves an initial investment of $100 million at time 0.
- In years 1 and 2, EBIT is expected to be $5 and $8 million.
- In years 1 and 2, depreciation is expected to be $3 million (each year).
- Starting in year 3, FCFF are expected to continue to grow at a rate of 2% in perpetuity.

Financial Statistics

- Risk-free rate = 4%
- Market risk premium = 7%
- Equity beta = 2
- The project will have the same mix of financing as the firm as a whole: $40\% = \frac{D}{E+D}$

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- Cost of debt = 5%
- Marginal corporate tax rate = 20%
- The firm is overall profitable and pays an effective tax rate of 7%

a. What is the firm's cost of equity? (3 points)

b. What is the firm's WACC? (3 points)

c. What are FCFF from years zero to two? (4 points)

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d. What is the NPV of the FCFF from years zero to two? (3 points)

e. What is the NPV of the terminal value of the project? (3 points)

f. What is the NPV of the project? Would you advise Ditta to proceed with the project? (2 points)

g. If Ditta could choose to abandon the project after two years, would this real option increase the NPV of the project? Briefly explain (1-2 sentences). (2 points)

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Problem 2: Lumon M&A (15 points)

Medtronic is considering acquiring Lumon Industries, a fast-growing biotech startup with cutting-edge neuro-modulation technology and proprietary memory partitioning patents. Lumon is still ramping up but has shown promising potential for growth. You are tasked with assisting in evaluating the acquisition using a comparable firm analysis. The acquisition is expected to generate $30 million in synergies.

	NeuroNova	Synaptech	Lumon
Net income	$8 mil.	$6 mil.	
Book equity	$24 mil.	$20 mil.	
Share price	$30	$36	
Shares	4 mil.	3 mil.	
Projected net income			$4 mil.
Projected book equity			$15 mil.

a. What is the average price-to-earnings (PE) and price-to-book (PB) ratio across the two comparable firms? (3 points)

b. What is the implied value of Lumon's equity under each multiple? (2 points)

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c. Based on the information available here, which multiple is likely more appropriate for valuing Lumon? Briefly (1-2 sentences) explain why. (2 points)

d. Using the value implied by your preferred multiple, what is the takeover premium (in millions of dollars) if Medtronic offers $80 million in cash for Lumon. (3 points)

e. Under the $80 million cash offer, what is the division of synergies in terms of millions of dollars? (2 points)

f. Suppose that after the announcement (but prior to the deal occurring), Medtronic's stock falls by 4% and Lumon's stock rises by 20%. Is this reaction unusual? Briefly (1-2 sentences) explain what this suggests about how investors view the deal. (3 points)

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Problem 3: Recapitalization at Globex (20 points)

Globex Corporation, led by charismatic CEO Hank Scorpio, is best known for its innovations in defense technology, energy beams, and ergonomic flamethrowers. Globex is currently a 100% equity-financed firm with 200 million shares outstanding, trading at $15 per share. Globex holds no cash.

Hank Scorpio announces that Globex will issue $1.2 billion in debt and use the proceeds to repurchase shares. Their cost of debt is 6% and their cost of equity is 13%. The firm faces a marginal corporate tax rate of 35%. After the recapitalization takes place, Globex's equity market capitalization falls to $2.5 billion.

a. What is Globex's enterprise value (EV) before the recapitalization? (2 points)

b. If Globex faced perfect capital markets, what would have been Globex's market capitalization and EV after recapitalization? (2 points)

c. Based on the actual market response, what is Globex's actual EV after the recapitalization? (2 points)

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d. Assume that Globex's debt level will remain at $1.2 billion going forward, and that their cost of debt and corporate tax rate will remain constant. What is the present value of the tax shield after the recapitalization? (3 points)

e. Based on the stock market reaction, what is the present value of other financial side effects associated with the recapitalization? (3 points)

f. Based on what you found above, which of the following trade-offs are the most plausible explanation for the additional side effects: financial distress costs, risk-shifting by management, risk aversion by management, or the money-to-burn problem? Briefly (1 sentence) explain how you reached this conclusion. (3 points)

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g. Briefly (1-2 sentences) explain how a trade-off you identified in (f) influences the relationship between leverage and firm value. (2 points)

h. After the recapitalization Hank Scorpio tells you "We should keep adding more debt—the cost of debt is lower than the cost of equity, so it'll reduce our WACC!" Even if capital markets were perfect, would his logic be flawed? Briefly explain (1-2 sentences). (3 points)

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Problem 4: Hedging Corn (22 points)

Prairie Valley Farms expects to sell 100,000 bushels of corn in 6 months. Concerned about falling corn prices, the farm is evaluating hedging strategies using corn futures and/or options. For simplicity, assume that corn itself (rather than corn futures) is the underlying asset for the options.

The spot price of corn today is $5.00 per bushel. The annual risk-free free rate is 5.5% and the annualized storage cost of corn is 10.60%. Note: for the options questions, you can ignore the storage cost.

a. What is the 6-month futures price of corn? Round your answer to two decimals places. (3 points)

b. Prairie Valley wants to hedge against a **decline** in corn prices. For both (i) corn futures and (ii) corn put options, indicate whether Prairie Valley would take a long or short position in the derivative to hedge this risk. (2 points)

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c. Draw a payoff diagram for *each* of the two strategies you identified above (i.e., two separate graphs). Label the axes and intercepts on your graphs, and clearly indicate which graph relates to which instrument. Suppose that the exercise price of the put option equals the futures price. (6 points)

d. Describe a portfolio containing both a call and a put that replicates the payoff of the futures strategy that you posited above. (3 points)

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e. 6-month corn puts with an exercise price equal to the futures price are currently trading for $1.05. What is the price of a 6-month call option on corn (with the same exercise price) implied by the put-call parity? Round your answer to two decimal places. (3 points)

f. Suppose that a 6-month call option with the same exercise price and maturity as in the previous question is currently trading for $0.65. Is there an arbitrage opportunity? If so, what trades could one make to obtain an arbitrage profit? (3 points)

g. What are two examples of real-world *limits to arbitrage* that could limit the ability of an arbitrageur to take advantage of such mispricing? (2 points)

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